2. Sedimentation and Absorption

The voids between the sand grains act as minute sedimentation basins. The colloidal matters arrested in these voids is in a gelatinous mass. Therefore, it attracts other finer particles by absorption.

3. Biological Metabolism

It is the growth and life process of living cells. When bacterias are caught in the voids of sand grains, the surface layer gets coated with a biological film. This film contains large colonies of living bacterias. They feed on the organic impurities present in water. They convert such impurities into simple harmless compounds by biochemical action. This results in the filtration of water.

4. Electrolytic Changes

The sand particles of filter media and ionized matter in water carry electrical charges of opposite nature. Hence, they attract each other and neutralize the charge of each other. This results in the alteration of chemical characteristics of water.

Types of filters

The filters are mainly classified as

1. Slow sand filters
2. Rapid sand filters
3. Pressure filters

The terms rapid and pressure represent high rate of filtration and slow indicate low rate of filtration.

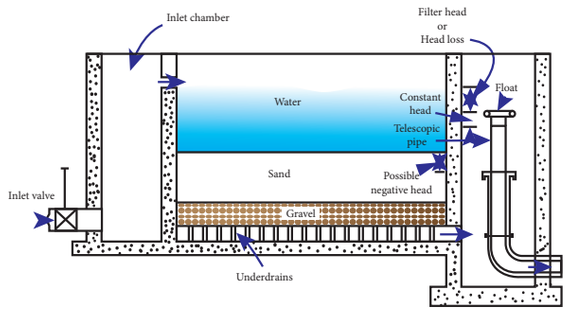
1. Slow sand filters

A slow sand filter is a water tight tank of 2.5 to 3.5 m in depth. It has a sand bed of 1 to 1.5 m thick, supported by a 0.3 to 0.75 m thick layer of graded gravel or broken stone (25 to 50 mm size), laid in layers. Beneath this, an under-drainage system consisting of open-jointed drains is laid over a concrete bed sloping towards a central longitudinal drain. The filtration is effected by gravity.

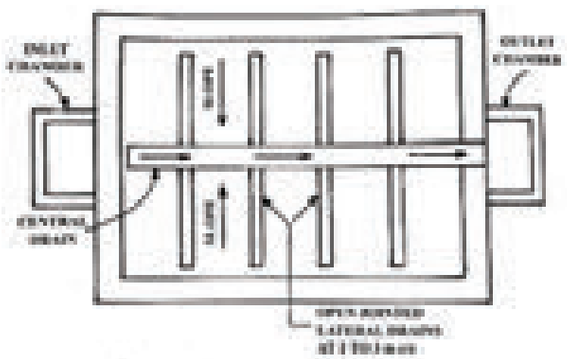
The rate of filtration is 100 to 200 lits/m^2/hour. It's bacterial efficiency is 98 to 99%. The filter bed is cleaned by scraping. This type of filter is unsuitable for waters having turbidity more than 50 ppm.

A typical plan and longitudinal section of a slow sand filter is shown in fig. It consists of the following essential parts.

i. Enclosure tank
ii. Filter media
iii. Base material
iv. Under-drainage system
v. Inlet and outlet arrangement
vi. Other appurtenances



Section of Slow Sand Filter



Plan of Slow Sand Filter

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